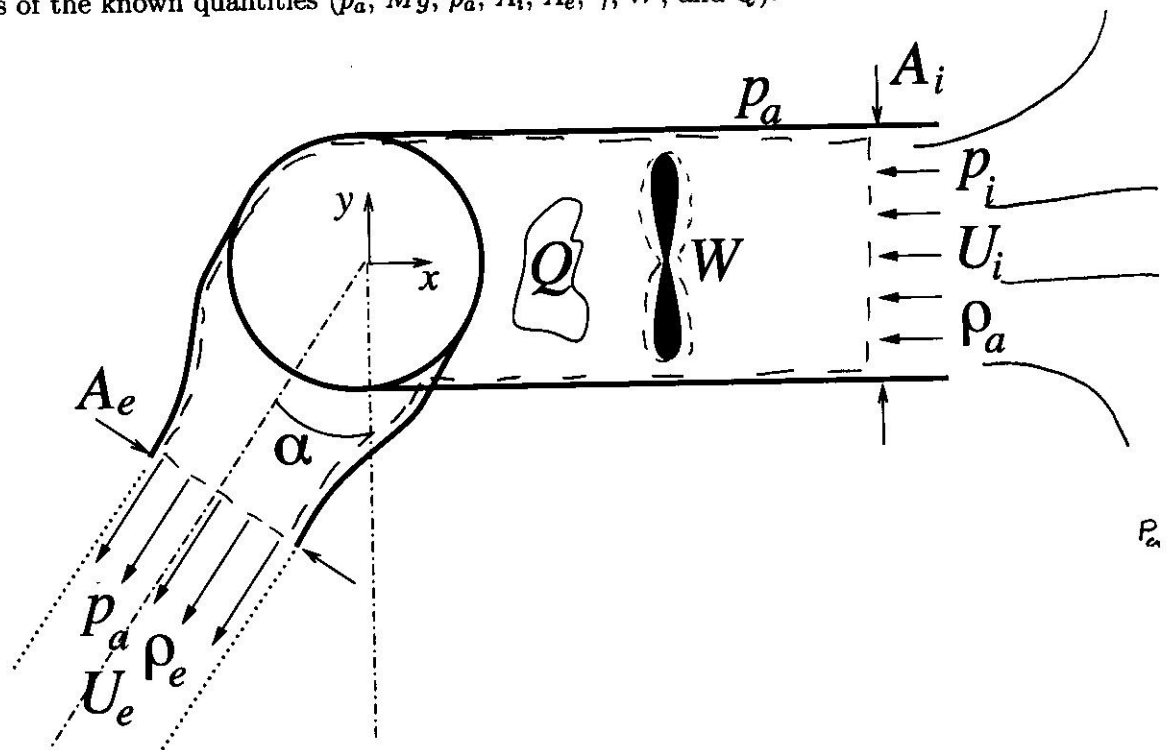


CL 11

The propulsion device of a VTOL (vertical take-off and landing) aircraft includes a horizontal intake of cross section A_i and a thrust vectoring nozzle of variable angle α and exit cross section A_e . When the aircraft is hovering (suspended at a fixed point), the engine must provide a force $\vec{F} = F_x \vec{e}_x + F_y \vec{e}_y$, such that the lift F_y equals the weight of the aircraft (Mg), while the thrust F_x is identically zero. The work per unit time on the air stream through the mobile blades is W and the amount of heat added per unit time is Q (including heat losses through the walls). For the analysis of the problem, one may introduce the following assumptions regarding the air motion:

- The effect of gravity is entirely negligible.
- Flow properties at the entrance and exit sections are uniform
- The air around the engine casing is at rest with uniform pressure p_a
- Upstream from the inlet the flow is inviscid and with constant density, such that $\rho_i = \rho_a$ and one may use Bernoulli's equation along any streamline to relate the ambient pressure p_a with that found at the entrance p_i .

With the above simplifications, write the equations that determine the values of U_i , U_e , p_i , ρ_e and α in terms of the known quantities (p_a , Mg , ρ_a , A_i , A_e , γ , W , and Q).



CONTINUITY:

$$\rho_a U_i A_i = \rho_e U_e A_e$$

ENTRANCE REGION:

$$p_a = p_i + \frac{1}{2} \rho_a U_i^2$$

MOMENTUM:

$$\rho_a U_i^2 A_i \vec{e}_x - \rho_e U_e^2 A_e (\sin \alpha \vec{e}_x + \cos \alpha \vec{e}_y) = -(p_i - p_a) A_i \vec{e}_x - F_x \vec{e}_x - F_y \vec{e}_y$$

$$X\text{-COMPONENT } (F_x = 0) \Rightarrow U_i = 2 U_e \sin \alpha$$

$$Y\text{-COMPONENT } (F_y = Mg) \Rightarrow Mg = \rho_e U_e^2 A_e \cos \alpha = \rho_a U_i^2 A_i \frac{\cos \alpha}{2 \sin \alpha}$$

ENERGY:

$$\rho_a U_i A_i \left[\left(\frac{p_i}{\rho_a} + \frac{U_i^2}{2} \right) - \left(\frac{p_e}{\rho_a} + \frac{U_e^2}{2} \right) \right] = W + Q$$